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Area 1: Sustaining Fracturing Operations Within a Deeply Mature Basin

F. K. Al Battashi, MEDCO Oman LLC; M. Rylance, IXL Oilfield Consulting

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Abstract

The heavily mature Area 1 cluster in the Southern Region of the Sultanate of Oman are generally characterized by low to medium permeability, demonstrate significant heterogeneity, suffer extensive depletion and contain medium through extremely heavy crude oil. Extensive hydraulic fracturing operations have managed to sustain and increase production, even in such a complex suite of fields but only through the extensive application of a Continuous Improvement (CI) programme, with line of sight to economics and value cut-off at all stages.

The ongoing campaign, performed for almost 15 years now, has demonstrated that while successful multi-year campaigns can be efficiently delivered, that this has only been achieved through incremental technical gains and improvements in the areas of candidate selection, fracturing implementation and technology selection. The foundational consideration is the candidate selection process itself, which is then supported by detailed well intervention preparation and of course the technical selection process for the fracturing placement itself. Many significant challenges have been incrementally overcome, in order to sustain the post-frac success and deliverability, including cement integrity and remediation, perforation quality and enhancement, tortuosity management, deviated wellbore candidate preparation, water-contact assessment and subsequent avoidance, fracture conductivity delivery as well as post fracture flowback management.

A Continuous Improvement (CI) program, which was performed year on year, has resulted in the ability to sustain effective hydraulic fracturing delivery even as the candidate pool has dwindled and the opportunity set has incrementally reduced. A very detailed process was established and updated with lessons learned and insights garnered with each new Annual campaign. This approach allowed for the hydraulic fracturing benefits to be delivered with an enduring success rate; and with an increasing capability to accept and develop poorer candidates. In order to enhance the post-frac production delivery, additional surveillance was performed with a particular view to maximising the fracture placement within the reservoir. The detailed fracture design was then adjusted, based on a range of inputs; and subsequently implemented. These new adjustments included solids stages; designed to constrain fracture height growth and this yielded a measurable improvement in the fracture geometries placed. A range of other incremental enhancements were made, which contributed to the ongoing success.

The paper follows the progress of the hydraulic fracturing programme across all of the AREA 1 fields, over the last fifteen years, as it has been technically progressed, improved and enhanced throughout the

selection, preparation, execution and delivery. The key choices and functional decision demonstrate how success van be sustained with a range of well types, which is a fitting summary as the programme now moves into the fracturing of horizontal wells.

Introduction

The AREA 1 cluster is a significantly complex, highly mature, brownfield group of small fields, see Figure 1, which has been undergoing production and development for some 30 years (as of 2025). Consisting of approximately 18 principal fields (of which 12 are currently under production). The concession has been operated by MEDCO Oman LLC since 2006, was extended in 2016 and is currently licensed to MEDCO until 2040. The fields consist of a number of differing reservoirs, across a range of complex stratigraphic horizons, containing moderate to highly viscous oil (μ) ranging from 10 cP to 850 cP. Depositional environments encountered vary but the majority can be described by fluvial, fluvial deltaic and glacial Palaeozoic sandstones.

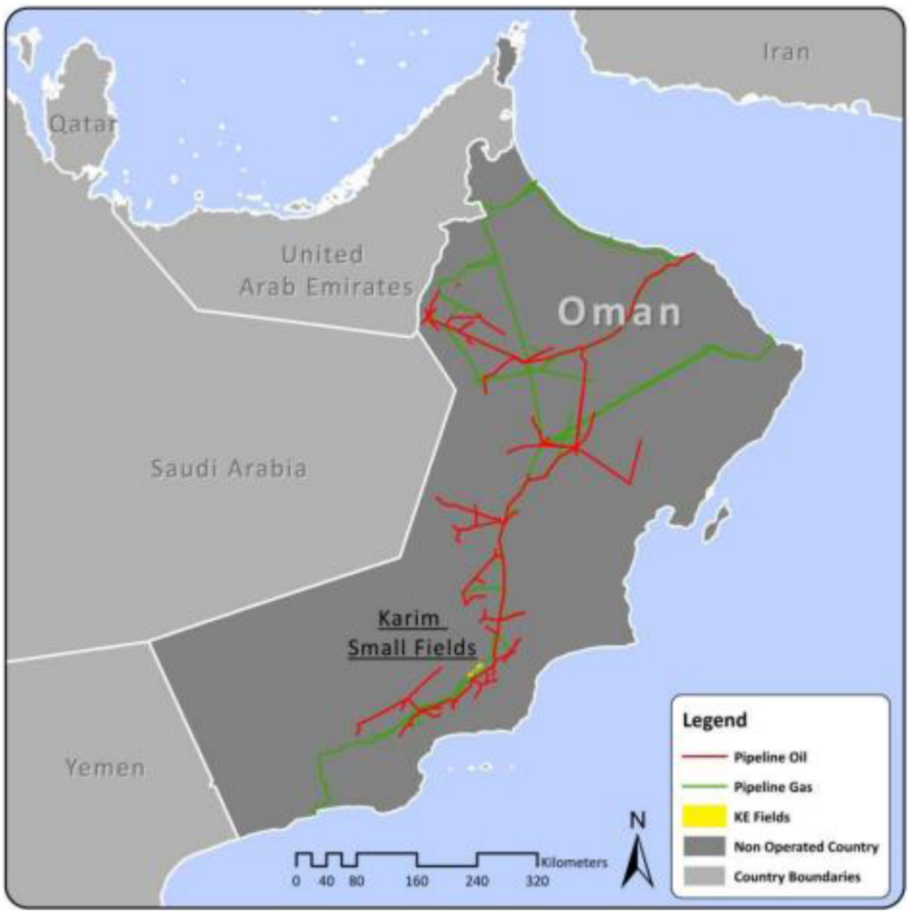

Figure 1—Location of Area 1 Cluster in Southern Oman.

Upon being granted the operatorship of the fields, MEDCO immediately planned and implemented a broad program to increase the deliverability from the AREA 1 cluster, through a selective suite of campaigns, which have been refreshed and reviewed on an Annual basis. These campaigns have included full blown field redevelopments, targeted infill drilling, recommissioning workovers, pilot EOR deployments and the consideration of hydraulic fracturing operations. Hydraulic fracturing was considered as a generally effective mechanism for a number of the challenges that existed within AREA 1, including removing drilling/production skin damage within the existing well stock, enhancing tighter formations, developing higher viscosities and uplifting the daily volumes of liquids being produced in order to improve the overall economics.

The results of these various intervention campaigns have been reasonably successful, with a significant production uplift having been achieved and sustained over a number of annual campaign programs. However, as with all mature field redevelopment plans and programs, the nature, the sequence and the impact of these operations change and reduce (with time), progressing in various phases from initial reactivation, consideration of artificial-lift techniques, through new infill wells, various forms of Enhanced Oil Recovery (EOR) and also implementation of horizontal or multi-lateral well drilling. The majority of the AREA 1 Fields consist of a suite of sandstone depositions, many of these sands are of fairly low permeability and also contain oil of variable quality, much of which is highly viscous. Therefore, it was natural that stimulation should form part of the consideration for field rejuvenation, and propped hydraulic fracturing was selected as one of the cornerstones of the redevelopment process. The remainder of this paper follows the various stages that were followed during the deployment of propped hydraulic fracturing as a solution on the AREA 1 fields.

Fracturing Program

The following sections will describe the onset and progression of the hydraulic fracturing program at MEDCO AREA 1, with particular emphasis on the technical challenges that were encountered and how these were managed on an annual campaign basis. Largely driven by intervention economics and deliverability; the program manages to continue even as candidate quality progressively deteriorates.

Historical Timeline

Hydraulic fracturing operations on the AREA 1 fields, under MEDCO operatorship, have been on a 15-year journey with the first fracturing operation taking place in 2010, and building slowly, through targeted learning to the present day, see Figure 2. Originally the candidates were dominated by the Haradh and Karim formations (Mirza et al., 2021), in more recent years additional candidates have been drawn from ever more challenging formations such as the Al-Khalata and Khaleel.

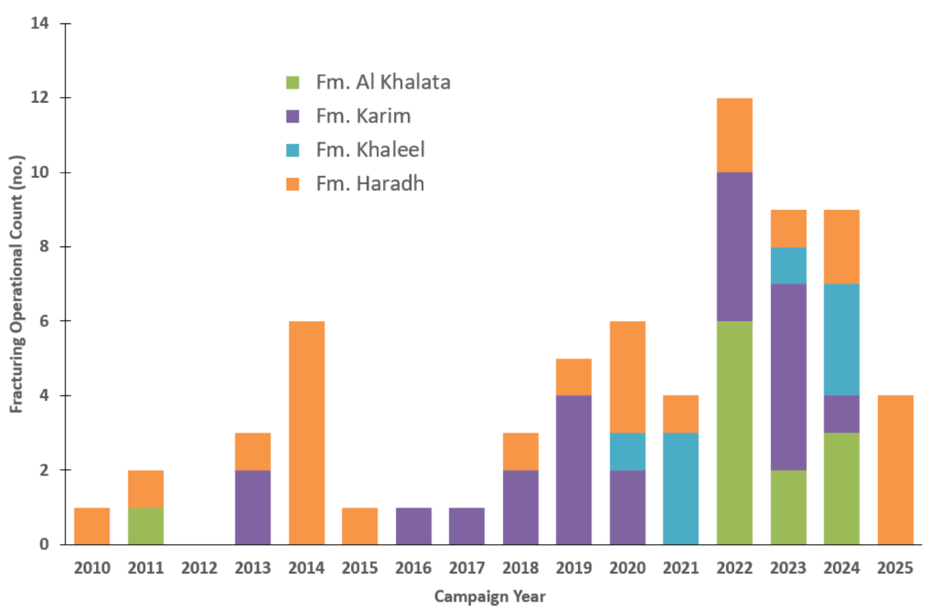

Figure 2—Fracturing Annual Operational Count on Different Formations.

As noted earlier the fracturing program at AREA 1 has been allowed to develop at a reasonable pace and with a consistent rhythm, with the Annual plan (Mirza *et al.*, 2010) and operational budget providing a natural break and recycling phase. Program execution takes place, whenever possible, within the first half of the year; allowing the second half to review current learning, consider the next candidates, prepare associated technologies and establish an execution plan for the next campaign. Quite often, in Oil & Gas, such a paced approach is not possible, particularly under Development, however for mature field rejuvenation it is considered to be an essential characteristic contributing to success.

In order to effectively achieve this continuous and successful, campaign, the approach that was taken was to develop the fracturing program in distinct phases based and driven entirely upon the available candidate pool (on an Annual basis). At the onset this was, of course, very much dominated by existing well-stock using typical candidate selection approaches (Martin & Economides, 2010) and (Mehrgini *et al.*, 2014), across all the fields within AREA 1. These existing wells contained a number of differing challenges, as would be expected with older wells not specifically designed for hydraulic fracturing operations. Such challenges would include:

i. Wellbore Deviation and Azimuth across (as often wells might be drilled from multi-well pads to minimize footprint),
ii. Casing/Wellhead design, casing-size/weight/grade, liner-laps, DV collars, corrosion and associated burst strengths,
iii. Zonal isolation integrity (poor cement bond across between, above and below any of the layers under consideration,
iv. Lack of intra-zonal isolation, as many of these wells had produced sequentially (from multiple layers) and finally,
v. Access issues, due to fish/junk, CIBPs, cement-plugs and other general production and WO related debris.

Many of these aspects were addressed within the initial campaigns, building a library of knowledge with regard to existing well status, which could be applied to future new wells. As the program progressed all of these issues were encountered and addressed, many of which will be described throughout sections of this paper.

Candidate Selection

The workflow initiated by the fundamental process of candidate selection, see Table 1, began with calculating the remaining oil in place and reserves within the target well/reservoir along with available pressure records from the offset wells. Since the target reservoir typically contains both oil and water, identification of an Oil Water Contact (OWC) is considered an essential step in the candidate selection process; in order to evaluate the correct height growth allowed, determine the effective/relative risk and avoid crossing to the mobile water zones and losing an ability to produce the well.

The stimulation treatments targeted eight oil fields and five formations; the lithology was mainly sandstone with a rock permeability of 50 mD to 100 mD. The fracturing treatment usually consists of two principal stages, the mini (data) frac stage and main frac stage. The main focus for the oil well candidate treatments were generally suffering from very limited inflow performance behaviour, typically with 10% to 20% water cut and the oil viscosity ranging from 10 cP to 30 cP. The stimulation treatment sizes ranged from small 80 Klbs treatments to moderate sized operations of 300 KIbs.

Table 1—A Typical Candidate Selection Process at AREA 1.

	Importance							25%	15.0%	17.5%				22.5%	5.0%	5.0%	5.0%	5.0%	100%
Rank	Well	Formation	Action	Top Perf (m)	Bottom Perf)m)	OWC	Distance to Water calculated	Distance to water	Barriers	Res. quality	Avg. k (mD)	Avg. Por (%)	Current Q_o (m3/d)	Current WC (%)	Cement quality	Deviation at perf.	Frac growth into zone with no data / OWC	Oil Viscosity	Final Score
1	**Well#1**	AKP	Completed	1369	1379	1195	174	175	Good	Good	3834	26%	7.98	0.30	Good	19.40	No	Heavy Oil	92.5%
2	**Well#2**	AKP	Completed	1528	1538	ODT 1612	ODT	65	No	Good	190	24%	9.90	4.72	Moderate	1.00	No	Light Oil	81.3%
3	**Well#3**	Haradh	Pending	1475	1485	1530	45	45	No	Good	330	21%	15.41	36.85	Moderate	0.00	No	Light Oil	71.7%
4	**Well#4**	Haradh	Cancelled	1484	1494	1534	40	40	No	Moderate	137	22%	10.59	26.29	??	1.00	No	Light Oil	67.1%
5	**Well#5**	AKP	Completed DF	1213	1223	1014	-	23	Some	Good	533	15%	4.59	21.49	Good	1.32	No	Heavy Oil	64.0%
6	**Well#6**	AKP-5U	Under review	987	997	ODT 1020	ODT	23	Some	Good	1395	25%	5.08	4.00	NA	??	Yes	Heavy Oil	63.9%
7	**Well#7**	AKP	Under review	1100	1110	1130	20	20	Some	Bad	????	20%	8.08	6.00	Good	1.70	Yes	Light Oil	63.8%
8	**Well#8**	AKP	Under review	1169	1179	???	ODT	13	Some	Good	1263	17%	5.30	1.47	Good	0.10	Yes	Heavy Oil	63.7%
9	**Well#9**	AKP-5U	Under review	1200	1210	984	30	30	Some	Moderate	175	13%	4.08	17.03	NA	??	Yes	Heavy Oil	60.0%
10	**Well#10**	AKP	Under review	1517	1527	ODT 1581	ODT	15	No	Good	907	18%	7.62	20.55	Moderate	1.00	Yes	Light Oil	55.3%
11	**Well#11**	AKP	Under review	1105	1115	ODT 1135	ODT	20	No	Moderate	145	21%	12.46	18.00	Moderate	1.00	Yes	Light Oil	54.0%
12	**Well#12**	Karim	Under review	1874	1884	ARA 1934	50	15	Good	Bad	5.5	15%	8.90	80.00	NA	0.00	No	Light Oil	52.0%
13	**Well#13**	AKP-5U	Under review	1946	1956	1984	28	28	Some	Moderate	12.7	14%	6.40	82.2	Moderate	1.40	Yes	Light Oil	51.0%
14	**Well#14**	Karim	Under review	1886	1896	ARA 1911	ODT	24	Some	Bad	0.44	11%	7.00	74.00	Good	1.00	Yes	Light Oil	49.0%
15	**Well#15**	AKP	Under review	1575	1585	ODT 1598	ODT	13	No	Moderate	134	22%	5.17	28.40	Moderate	26.75	Yes	Light Oil	44.9%
16	**Well#16**	MG	Under review	1007	1017	1022	5	5	No	Good	3478	25%	3.12	26.00	Good	14.60	Yes	Heavy Oil	44.8%
17	**Well#17**	Mahwis	Under review	1313	1323	1337	14	14	No	Good	259	23%	8.14	77.00	Good	1.00	Yes	Light Oil	44.6%
18	**Well#18**	Karim	Under review	1996	2006	2005	-1	1	Good	Bad	7.6	8%	7.80	59.20	NA	8.90	Yes	Light Oil	43.6%

Wellbore Preparation

A well integrity assessment is absolutely vital prior to performing hydraulic fracturing operations, since the well will be under high bottom-hole pressure conditions during the fracturing operations. As standard practice, a cement bond log is almost always planned to be performed in order to check the current integrity of the cement behind the casing and initiate a remedial cement process should that be required. In addition, on some occasions casing leak identification is required in these relatively very old and single casing wells; in order to assess the potential requirement for other integrity operations, such as the setting of a casing patch and/or a dedicated scab liner assembly. Additionally, the well trajectory and orientation with regard to in-situ stress regime is also assessed (Hallam & Last, 1991), as deviated or highly inclined wells are susceptible to screening out due to near wellbore complexity and Near Well-Bore Friction (NWBF).

After selecting the proposed fracturing candidates, then the wells then undergo a pre-fracturing preparation procedure, where the necessary modifications are completed, such as re-perforation, cement squeeze, patching and ultimately installation of the fracturing packer and string (simple frac-string but usually with a BHP memory gauge). The next operational stage consists of the breakdown, SRT/SDT and mini (data) frac; where a pad of fracturing fluid is pumped into the formation in order to simulate the fracturing procedure and measuring the breakdown pressure, followed by a temperature log to assess the height growth and eventually modification of the main fracturing design. Lastly, the well is fractured and cleaned using Hoist and/or a Coiled Tubing Unit and an artificial lift system is installed.

While some near-wellbore Screen-Out (SO) had taken place (ca. 20%) this was exacerbated in 2023 (ca. 50%) and was determined to be related to a change of vendor and gun selection for the perforating for fracturing. An additional improvement to the preparation process was performed, verifying and enhancing the perforation gun selection/well combination. The new approach applied industry knowledge related to requirements such as Entry Hole Diameter (EHD) over depth of penetration, reducing perforation interval in inclined and poorly oriented wells, as well as perforating in a preferred orientation. This selection process resulted in a significant reduction in the NWBF, and an elimination of near-wellbore SO events to zero in 2024, see Figure 3.

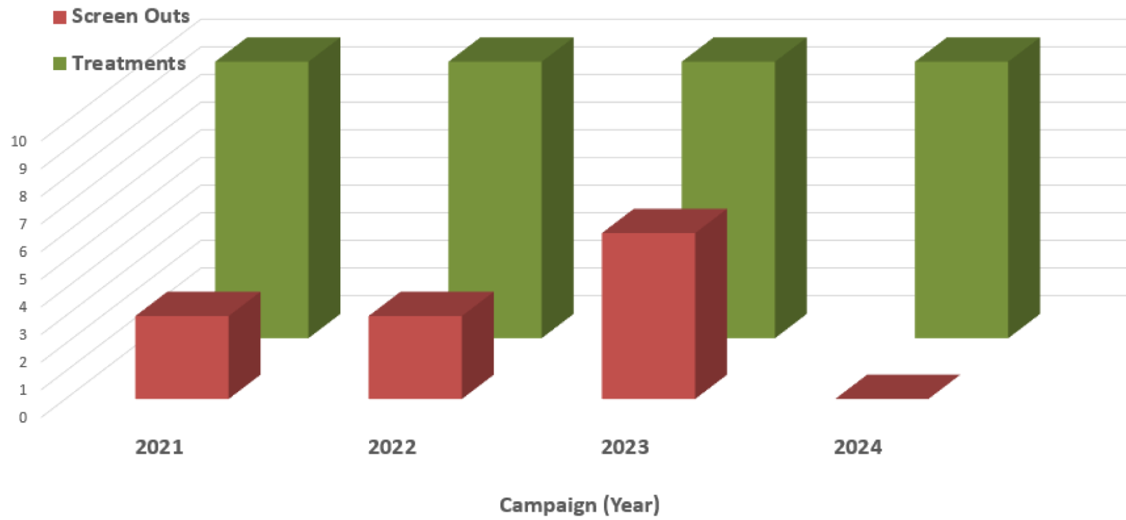

Figure 3—Screen-Out Incidence 2020 – 2024.

Proppant Flowback

The change in the perforation gun selection also assisted with enhancing the hydraulic fracturing design even further, the extensive EHD that the new gun achieved allowed consideration of new technology by adding new technologies such as Rod-Shaped Proppant (RSP). This extensive EHD allowed s mall stage of RSP (McDaniel *et al.*, 2010) to be pumped in the last stage of the fracturing treatment, which then dramatically reduced the proppant flow back and artificial lift system failure, assisting with economic optimization.

The advantage from injecting the Rod-Shaped Proppant is that the higher porosity, permeability and resulting conductivity, with the spatial packing of 0.65″ cylinders compared to 0.42″ spheres is substantial. In addition, the interlocking consolidated proppant pack (due to the cylindrical shape) resists the driving forces that result in proppant flowback. The RSP was implemented in two wells in the 2024 campaign and both wells continue to be actively producing without any sign of proppant flowback effect on the artificial lift system load and fluid level indicates that the proppant is not accumulating in the wellbore. Compared to the previously fractured wells shown in Figure 4 below, the number of hoist entries is significantly reduced from an average of three entries in the first six months to zero, substantially reducing the workover hoist cost.

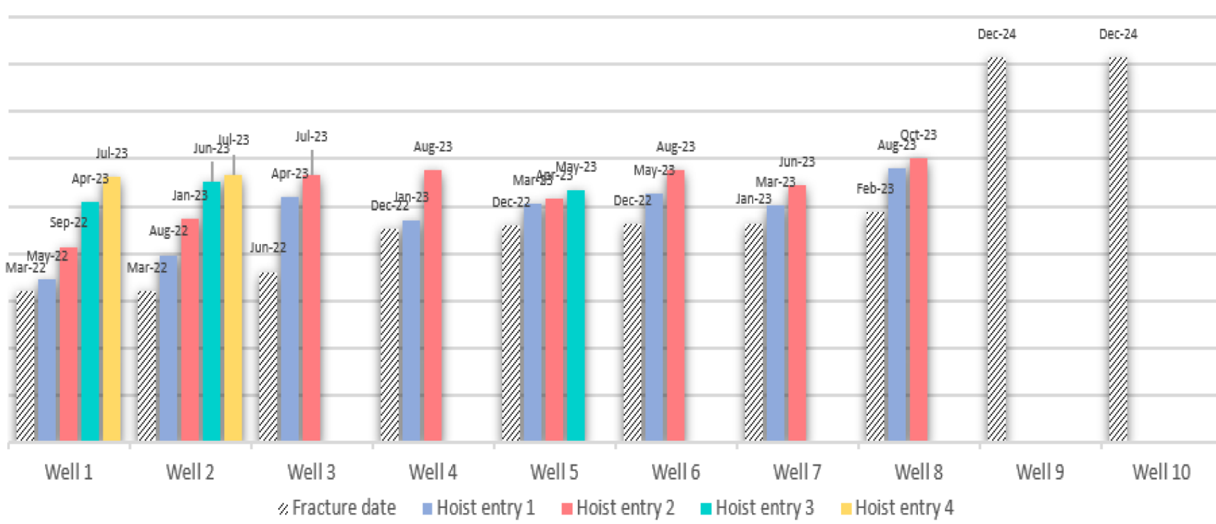

Figure 4—Hoist Entry Record in Fractured Wells (months without re-entry).

The use of the RSP has been extremely advantageous where it has been applied and this approach is now considered a reliable technology for dealing with the problems of proppant production. Whereas earlier trials with various Resin-Coated Proppant (RCP) materials had only provided partial and temporary relief, the sustainability of the RSP has now been well established.

Constraining Height

The majority of the fields within AREA 1 have an active OWC, which means that knowledge and control of fracture height generation is absolutely key, in order to ensure that the fracture treatments are not propagated into and below the water-contact. Fracture height control, and its specific control, is an area of detailed fracture design that has been investigated at depth and is very well summarized by Rylance & Korovaychuk (2023). The specific mechanisms that can be used to control fracture height growth are noted in Figure 5.

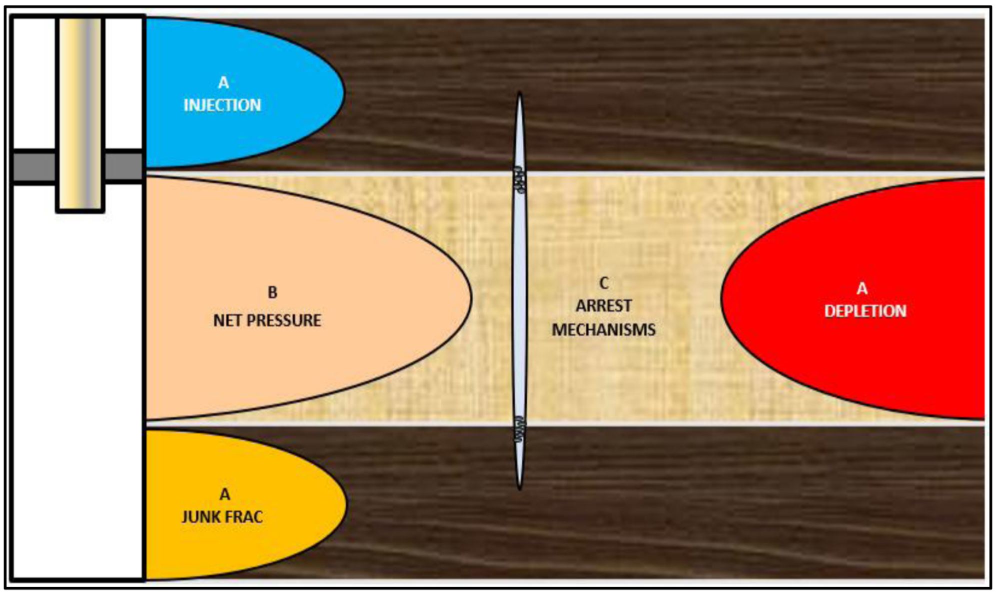

Figure 5—Mechanisms that can be Utilised to Control Fracture Height -Growth (after Rylance & Korovaychuk, 2023).

The three principal mechanisms, A, B and C, that can potentially be used to directly impact the magnitude of fracture height growth can be classified as (A) Effective-Stress Control, (B) Net-Pressure minimisation and (C) Physical blunting/arrest mechanisms. In the case of fracture height control using (C), then this mechanism can be implemented in a stage 'a priori' or during the execution of the main fracture stage.

This third approach is a form of bridging/blunting of the fracture growth through the application of engineered particulates that are strategically pumped (Nguyen & Larson, 1983) and (Larson & Nguyen, 1984). It has been noted that it is quite likely that this methodology had been occurring for several years (Miller & Warembourg, 1975), without a full and well-defined understanding of the means and way that the solids were physically impacting the fracture geometry behaviour. This is achieved for upper fracture height arrest, by using solids that are introduced to the slurry schedule after the pad creation of the body of the frac and used to progressively 'screen-out' the fracture along the fracture upper growth edge of the fracture. In a similar fashion, for avoidance of the OWC these dense solids can be used to create a similar effect along the lower fracture-edge, settling rapidly in a specialized stage prior to main fracturing to effectively plug the fracture from growing any lower.

For the hydraulic fracturing campaigns that were performed in 2023, 2024 and 2025, this technique was increasingly employed in those scenarios where downward growth into water was considered to be a major risk. This approach was to prove highly successful in arresting downward fracture growth, which was able to be demonstrated repeatedly through information garnered from the use of both temperature and

Radio-Active (RA) tracer logs. Ultimately of course the success of this approach, has been most effectively demonstrated through the continued water-free and low water-cut production form the candidate wells in their post-frac phase.

Fracturing Delivery

A total of sixty-six hydraulic fracturing stimulation operations have been conducted within the KSF structures, with twenty-seven wells fractured in the Karim formation alone and twenty-one within the Haradh formation, see Table 2. Post fracture production assessment was as follows:

- *Exceed Expectations:* Post fracture production assessment has shown that forty of these fracturing operations resulted in both a highly satisfactory initial oil rate along with a sustained production over an extensive five-year period.
- *Deliver Expectations:* Post fracture production assessment has shown that seventeen of these fracturing operations resulted in a satisfactory initial oil-rate but with a shorter production cycle and sustained treatment lifespan.
- *Fail to Meet Expectations:* Post fracture production assessment has shown that nine of the treatments resulted in delivering wells that underperformed.

Table 2—Success Measure for total AREA 1 Hydraulic Fracturing Operational Experience.

Field	Formation	Total Operations	Over Delivery 100%	Acceptable Delivery 50%	Under Delivery 0%	Success Ratio
Field S	Haradh	10	9	1	0	95%
Field S	Karim	9	5	3	1	72%
Field J	Haradh	11	4	6	1	64%
Field J	Karim	5	1	2	2	40%
Field W	Karim	13	10	1	2	81%
Field I	Khaleel	4	2	1	1	63%
Field H1	AKP	1	0	1	0	50%
Field H2	AKP	1	1	0	0	100%
Field A	Al Shomou	1	0	0	1	-
Field A	AKP	1	1	0	0	100%
Field Z	AKP	8	5	2	1	75%
	Water Injectors	2	2			
	Total Jobs	**66**	**40**	**17**	**9**	**Avg. 75%**

The average success ratio achieved within the Haradh formation has ranged from 64% to 95%. On the other hand, the Karim formation was also fractured stimulated (Al-Kalbani *et al.*, 2018) in three specific fields, where the average success ratio achieved has ranged from a value of 40%, to 72% and 81% respectively. As an example, the oil gain in the fractured wells increased up to 274% in the J Field when specifically targeting the Haradh formation.

With an average success ratio of 75% for the entire program, this is considered to have been an exceptional achievement for such a mature suite of fields, with such challenging conditions; and continuing and sustaining such a track record over time will become increasingly difficult. With all of the underlying factors that challenge the hydraulic fracturing delivery, continuing to determine quality candidates from the old well stock will not be straightforward, and like all fracture campaigns appropriate economic cut-offs will need to be applied.

For the 50 wells that have been fracture stimulated within the AREA 1 field cluster, between the period 2020 and 2024, then the delivered value, in terms of annual impact and also as a consolidated intervention and enhancement program, can be summarised as shown in Figure 6 Left and Right:

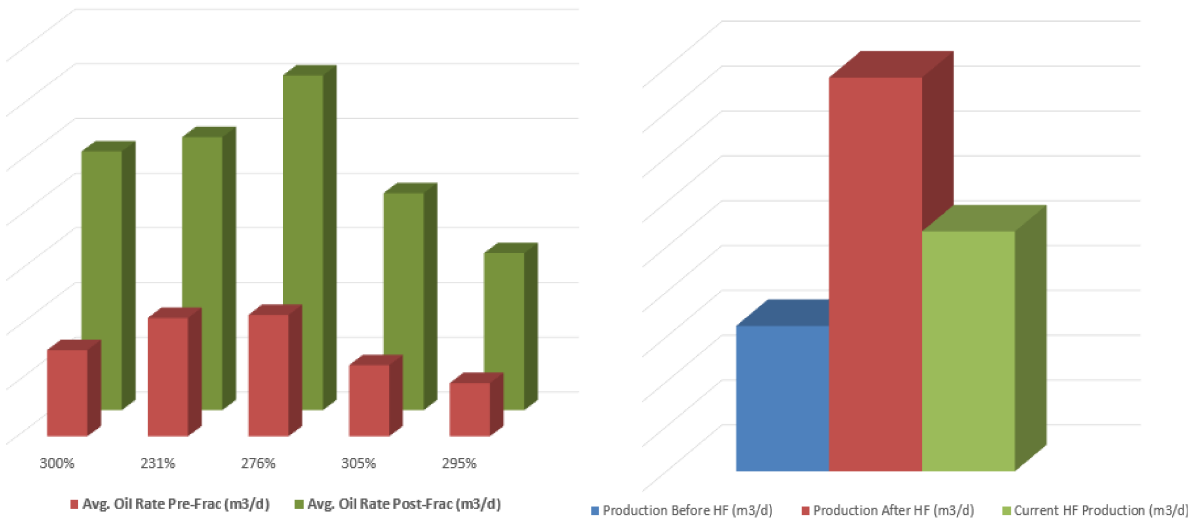

Figure 6—Pre- and Post- Frac Rate Comparison 2020-24 and Current Campaign.

Since the implementation of the hydraulic fracturing operational campaign at AREA 1, a total of seven individual fields has been selected, with four distinct target formations also being treated. The following chart, Figure 7, represents the average oil production enhancement that has been achieved for each of the different formations, within each of the fields, during the stimulation campaign period from 2020 - 2024.

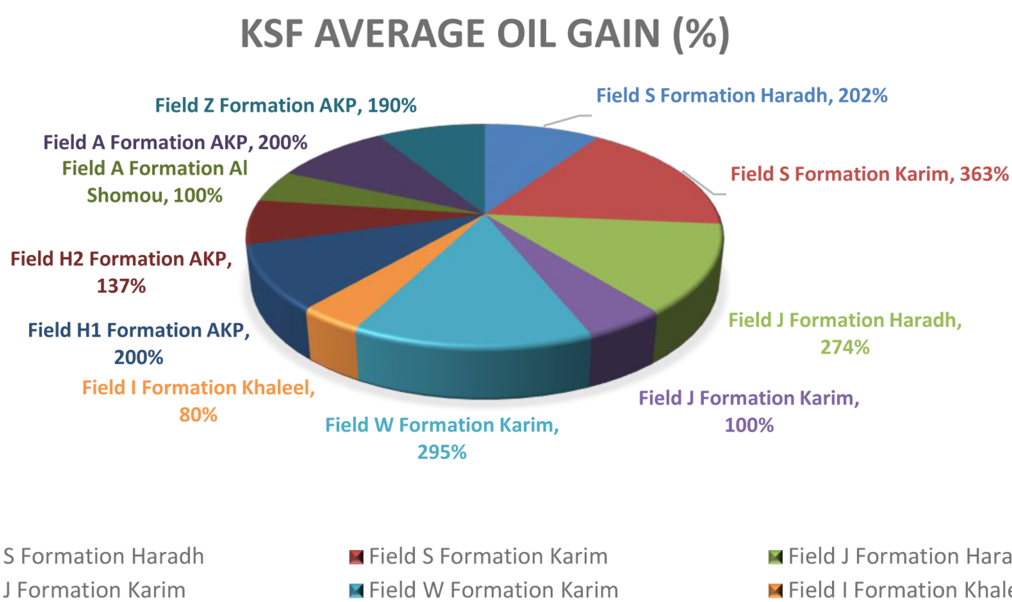

Figure 7—Average Production Increase by Field and Formation (2020 – 2024).

Clearly, the majority of treatments have taken place in the Haradh and Karim formations, as noted in Table 2, however as shown in Figure 2 recent years have seen the addition of Al-Khalata and Khaleel. It is expected that as the campaign progresses there will be more variance in the formation types treated, and while we are already achieving satisfactory results with Al Khalata, more progress needs to be made with formations such as Khaleel … etc.

Fracture Challenges

As the AREA 1 program advances and the fields continue to mature, the future of the hydraulic fracturing operations face several critical challenges, particularly due to the evolving nature of the reservoir development strategies. Many of the readily accessible, high-potential well candidates for fracturing have already been exploited, leading to a shift in focus towards the even more mature fields where primary recovery has already taken place. Increasingly, these fields are being managed through secondary and tertiary recovery methods such as waterflooding, Enhanced Oil Recovery (EOR) techniques like steam flooding, and also Chemical EOR (CEOR) approaches including polymer flooding.

This transition can also be seen in AREA 1, as waterflood and steam flood projects have already been introduced to several formations, which present new complexities for the application of hydraulic fracturing, or eliminates it completely. This is due to the fact that the reservoir conditions, fluid saturations, and pressure regimes are significantly altered by these methods; and hydraulic fracturing is not always an appropriate option. Where applicable, then hydraulic fracturing in such environments requires highly tailored designs, deeper knowledge regarding reservoir characterization, and advanced monitoring technologies to ensure both effectiveness and compatibility with such ongoing EOR operations. Integrating hydraulic fracturing into those fields already under EOR or CEOR development remains a significant technical and economic challenge that MEDCO needs to address through further innovation and interdisciplinary collaboration.

Poor Mobility (k/μ)

Increasingly, the annual fracturing program is having to consider lower permeability (k) environments, containing more challengingly viscous hydrocarbon (μ), leading to a deteriorating mobility (k/μ) environment in which to stimulate. This will necessitate the adoption of new and more challenging stimulation techniques, which will directly address and consider these limiting factors and will result in the implementation of more unconventional approaches. While this will result in stimulation being achieved, it will also result in lower Energy Returned On Energy Invested (EROEI), i.e., stimulation cost in a ($/bbl) measure, will associatively increase.

However, in order to achieve this, then we can of course rely on significant learning from the Unconventional Resource delivery that has taken place, largely in North America, where intensive stimulation of extensive horizontal wells has resulted in economic access to challenging resources. Albeit, we will have to accept that the number of wells that will be available to us, in order to implement (and learn) the solution, will be far less than we have had access to in those NA assets. All the same, use of technologies such as slickwater or HVFR, local sand, cluster and staging, plug and perforation and other unconventional approaches will all likely become potential/realistic considerations.

Thin Oil-Rims

Even with the use of the height constraint methods, as noted above, increasingly there are an increasing number of opportunities where the allowable fracture height growth is so low as to make fractured verticals unrealistic; in that regardless of the use of bridging that growth outside the limited height allowed is almost guaranteed to take place. Under these conditions, the use of horizontal wells that penetrate the oil-rim, drilled in the direction of the S_{Hmax} stress and stimulated along their entire length is likely the most appropriate manner in which to consider these opportunities.

For such new horizontal wells, it is quite likely that a relatively simple approach should be taken to completion, which would consist of an open-hole system, with external isolation (swell elements or mechanical packers) and sliding-sleeve access to each of the segments. This would allow lowest cost, with maximum reservoir contact and straightforward isolation (of any segment) should water breakthrough occur. Under this scenario, fracturing would be longitudinal along the wellbore and due to the Kirsch effect would result in very extensive fracture half-lengths with very little height growth. These fractures and sliding sleeves could also be placed and operated with CT, making fracturing quick, slick and effective, all improving the overall economics.

Conclusions and Recommendations

The MEDCO AREA 1 hydraulic fracturing campaign has managed to remain effective and economic for a number of years now, even with a deteriorating candidate pool. Adaptability, flexibility, technology acceptance and innovation have all contributed to sustaining the fracturing program on an annual basis, and the challenge continues. A complex interchange of challenges and opportunities were faced, as production profiles thin and reservoir pressures decline, optimizing hydraulic fracturing tactics become essential for maintaining economic viability. This requires a move toward data-driven decision-making, innovative stimulation technologies, and tailored reservoir characterization. Coordination across disciplines, constant innovation is key to extending the productive lifetime of mature assets. Ultimately, success in such basins depends on balancing technical aspects with operational efficiency to unlock the remaining potential in a cost-effective and sustainable means.

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Nomenclature

k	= *Permeability*	*(mD)*
μ	= *Fluid viscosity*	*(cP)*
S_{HMax}	= *Maximum Horizontal Stress*	*(psi)*
BHP	= *Bottom-Hole Pressure*	*(psi)*
CEOR	= *Chemical Enhanced Oil recovery*	
CIBP	= *Cast Iron Bridge Plug*	
EHD	= *Entry Hole Diameter*	*(inches)*
EOR	= *Enhanced Oil Recovery*	
EROEI	= *Energy Returned On Energy Invested*	
HVFR	= *High Viscosity Friction Reducer*	
NA	= *North America*	
NWBF	= *Near Well Bore Friction*	
OWC	= *Oil Water Contact*	*(ft.TVD.SS)*
RA	= *Radio-Active*	
RCP	= *Resin Coated Proppant*	
RSP	= *Rod Shaped Proppant*	
SDT	= *Step-Down Test*	
SO	= *Screen-Out*	
SRT	= *Step-Rate Test*	
WO	= *Work-Over*	

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